Team 18 Final

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evaluation guideline

Relevance to practical applications in computer vision

Novelty, reasonableness and effectiveness

It’s okay to combine various popular models for an interesting task.

The presentation, clarity and accuracy of the video and the report.

It is essential to clarify which parts are your own contributions and novelty

Reference Paper :

Revisiting MAE pre-training for 3D medical image segmentation

Published on **CVPR 2025**

From 2D to 3D: Why Medical Imaging Is Different

Aspect	2D Image Tasks	3D Medical Imaging (CT/MRI)
Data Structure	Single image (H × W)	Volumetric cube (H × W × D)
Spatial Depth	Flat, no depth continuity	Strong depth continuity across slices
Resolution	256–1024 px	160^3–512^3 voxels（up to millions of points）
Visual Cues	Strong textures, colors, edges	Low contrast, subtle boundaries
Noise & Variability	Controlled, predictable	Scanner-dependent, noisy, heterogeneous
Annotation Cost	Cheap, fast (bounding boxes)	Expensive (3D segmentation by clinicians)
Class Imbalance	Moderate	Severe（lesions can be <1% of volume）
Model Difficulty	2D CNN/ViT work well	Requires 3D CNN with inductive bias

Why Transformers Underperform in 3D Medical Imaging

Challenges for 3D ViT / DETR models:

- Self-attention scales quadratically → GPU memory explosion
- Patch count extremely large in 3D volumes
- Requires huge datasets (not available in medical domain)
- Poor voxel-level precision
- No strong inductive bias for anatomical continuity

Result: CNNs (especially U-Net variants) consistently outperform Transformers in 3D segmentation.

Revisiting MAE: Original Idea

Masked Autoencoder (MAE) is a **self-supervised learning framework** where the model learns by reconstructing heavily masked inputs.

Core concept of MAE:

- Mask around 75% of patches
- Encoder sees only visible patches → Forces the model to infer global structure instead of memorizing details.
- Lightweight decoder reconstructs masked patches
- Strong for self-supervised pretraining
- Works extremely well for ViT in natural images.

Revisiting MAE: Original Idea

Masked Input (~75% Masked)

Reconstructed Output

Core Concept

- Mask around 75% of patches
- Encoder sees only visible patches → learns global structure
- Lightweight decoder reconstructs masked patches

Key Benefits

- Strong for self-supervised pretraining
- Works extremely well for ViT in natural images

Why MAE Fails on CNN (Without Fixes)

Problems when applying MAE to CNNs:

- CNNs are dense → masking destroys spatial continuity
- Feature maps collapse under high mask ratio
- Skip-connections in U-Net break under random masking
- CNNs cannot natively handle sparse inputs

S3D Innovation #1: Making CNNs Survive Masking

1. Sparse Convolution (SparseConv3D)

- Instead of treating masked voxels as zeros, **S3D models them as absent.**
- **Convolution is computed only on visible voxels**, preserving true neighborhoods.
- Prevents feature collapse under high masking ratio.

2. Sparse Normalization (SparseBatchNorm)

- **Traditional batch norm expects dense statistics.**
- SparseNorm computes mean/variance only from visible voxels.
- **Avoids unstable activations** when 75% of voxels are missing.

These steps make CNN encoder becomes mask-aware, just like ViT MAE. It no longer collapses when 75% of the input is gone.

S3D Innovation #2: Reconstructing Dense Volumes

3. Mask Token Densification (MTD)

- Transformer MAE inserts a trainable mask token.
- CNN cannot insert tokens into grids.
- S3D introduces **dense placeholders** for missing voxels.
- Ensures decoder receives *spatially aligned* feature maps.

4. Densification Conv (3×3×3)

- Converts **sparse encoder output** → **dense 3D feature map**.
- Local convolution fills gaps, producing a coherent representation.
- The decoder can now operate like a normal U-Net decoder.

S3D Innovation #3: Researching best masking ratio

5. Residual Encoder (nnU-Net)

- Strong CNN backbone, ensures stable gradients under sparse inputs.
- Natural fit for volumetric segmentation.

6. Optimal Masking Ratio: 75%

- Extensive ablations show highest downstream performance
- Lower ratios → not enough challenge
- Higher ratios → CNN collapses even with sparse conv

Outcome: S3D becomes the first MAE that works well on 3D CNNs, outperforming ViT-based 3DMAE and other SSL methods.

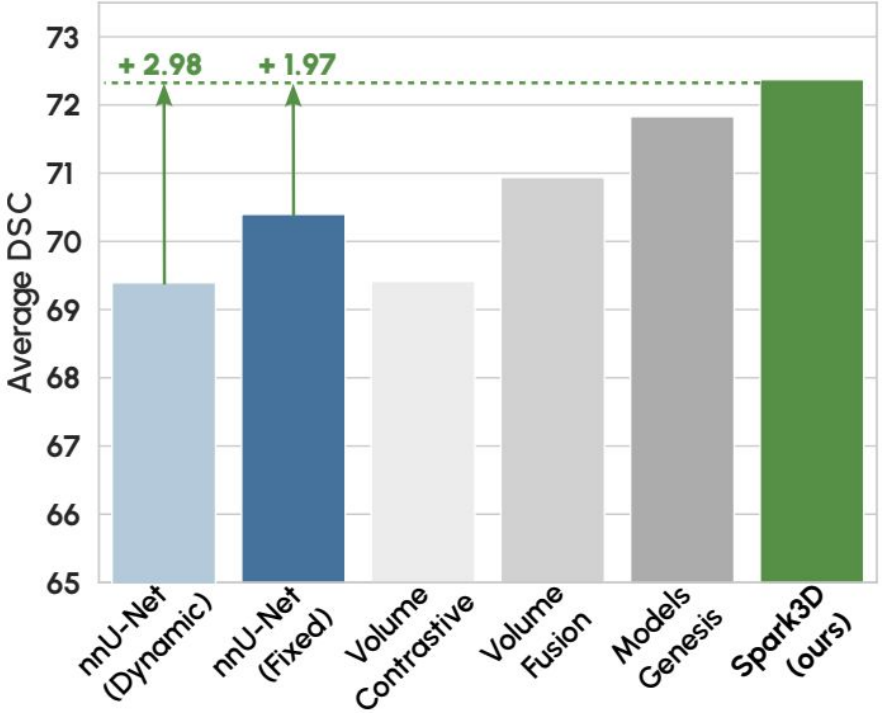


Figure 1. **Well-configured MAE pre-training for CNNs is state-of-the-art.** When comparing our MAE pre-trained model on all eleven test Datasets, Spark3D achieves almost +3 DSC points compared to the strong nnU-Net baseline and outperforms current SSL methods.

Why S3D Outperforms 3D MAE (ViT-based)

S3D advantages over ViT-based 3D MAE:

- Handles full-resolution 3D medical volumes
- Requires less computation
- Maintains anatomical continuity
- CNN inductive bias perfectly matches medical imaging
- Much more stable during fine-tuning
- Stronger Dice performance across multiple datasets

Applying S3D to Our Project

Task: Heart Valve Localization

- Volumetric cardiac CT
- Main hurdles: High spatial complexity, Small target structures, Limited annotated data

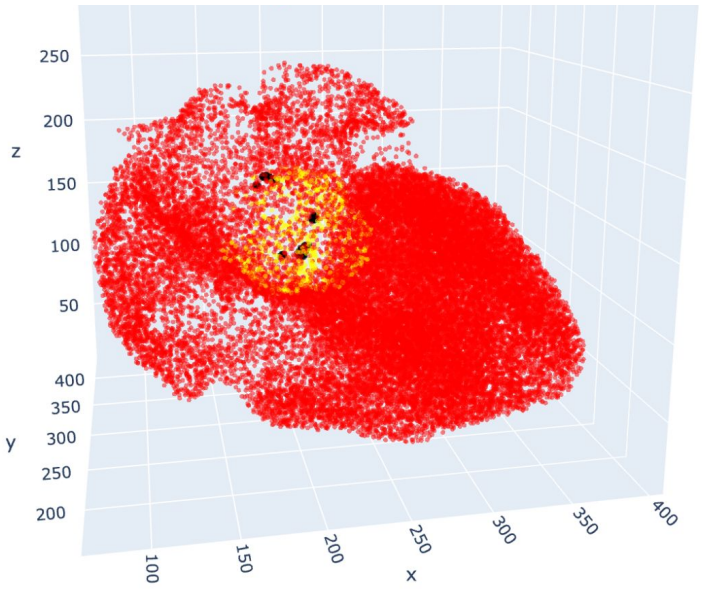
Why S3D Helps

- Strong global anatomical representation
- Handles 3D continuity
- Reduces need for labels via self-supervised pretraining
- Better downstream performance for detection and segmentation

AICUP 2025 (Link)
Segmentation For Heart CT

AICUP Introduction

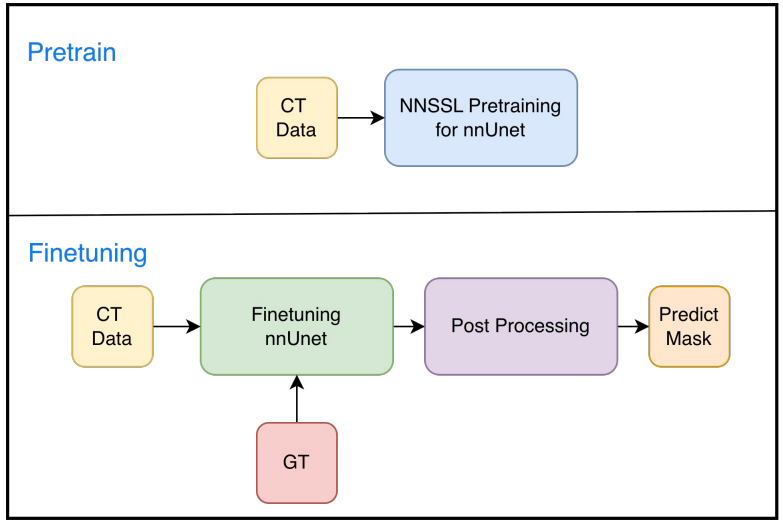
- 3D Segmentation Task
- Input: Heart CT Data
- class
  - 0: background
  - 1: heart muscle
  - 2: aortic valve
  - 3: Calcification
- Challenge
  - Long tail class for class 3



Method

Pretraining + Finetuning + Post processing

- Pretraining
  - nnssl S3D method
- Finetuning
- Post processing
  - Using 3*3*3 kernel to erase the isolated label.



Result

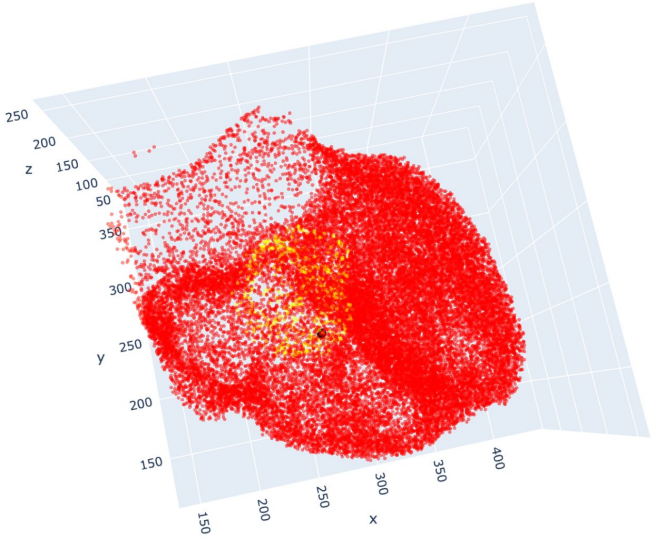
Results

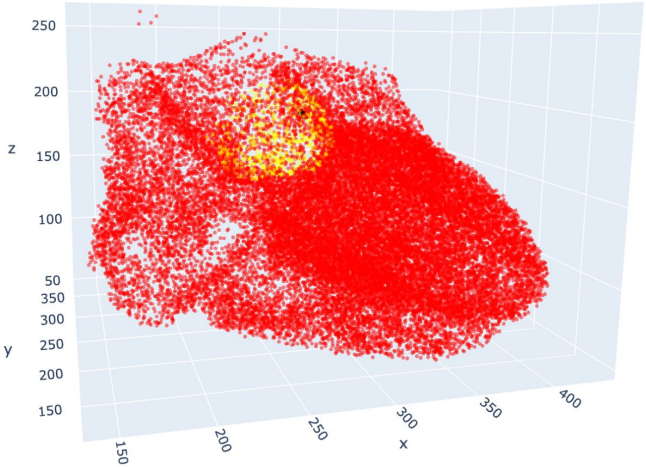
- Metric: Average of Dice and IoU
- Best score 79.678 %
- Private Rank: 56/568
- Public Rank: 36/568

	Score
nnUnet	0.79676
nnssl + nnUnet	0.79042
nnUnet + Post processing	0.79678

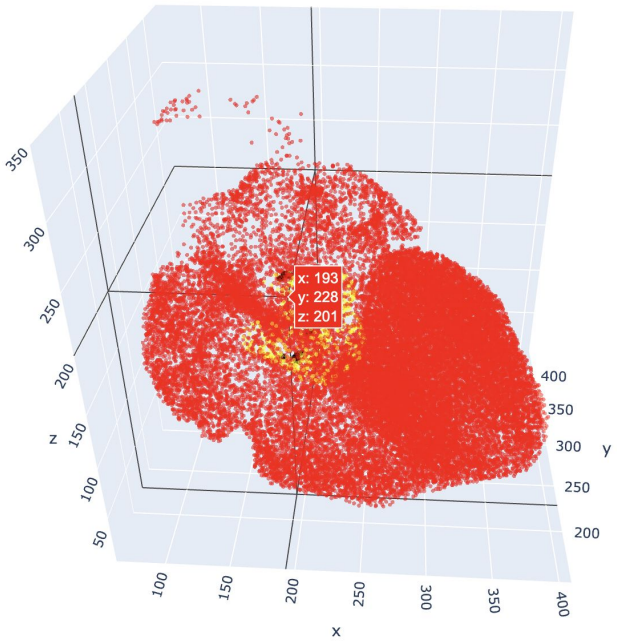
Demo

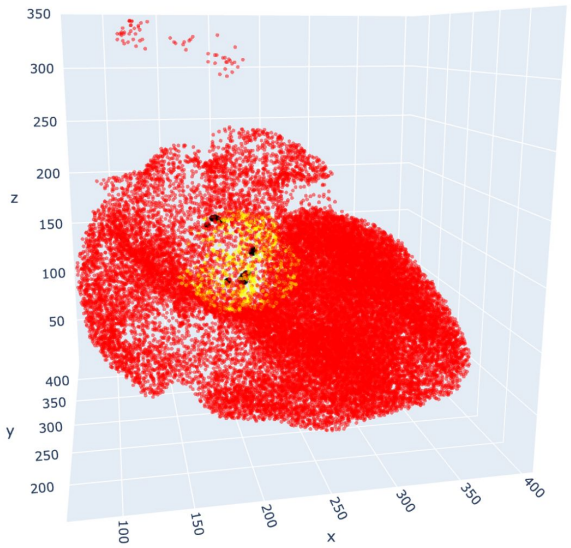
Demo Image – patient52



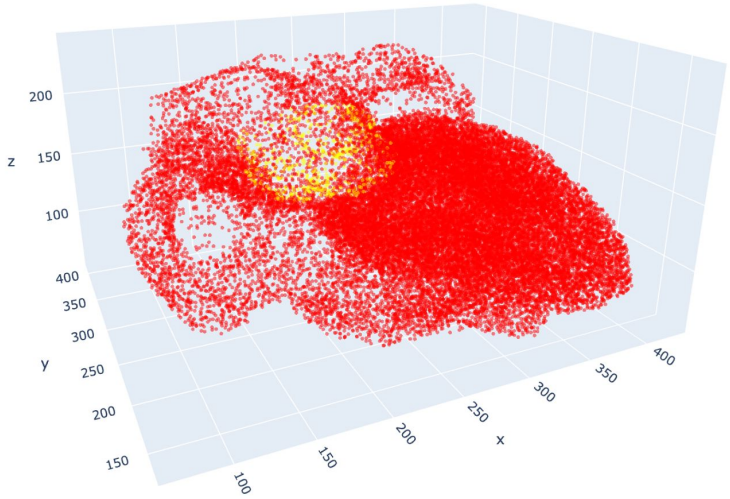


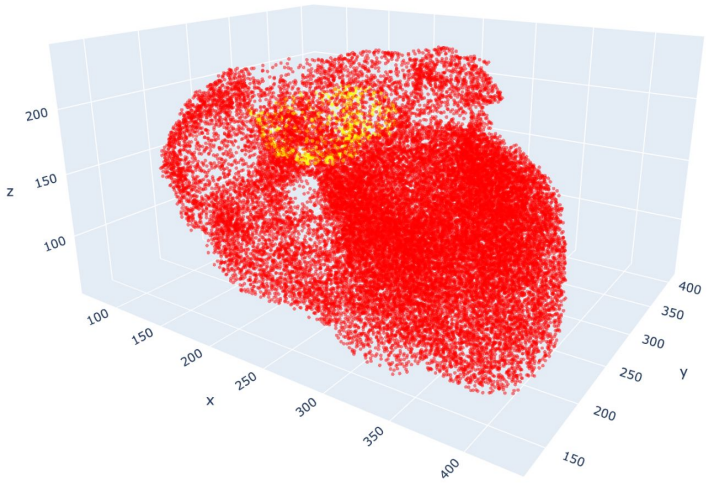
Demo Image – patient53



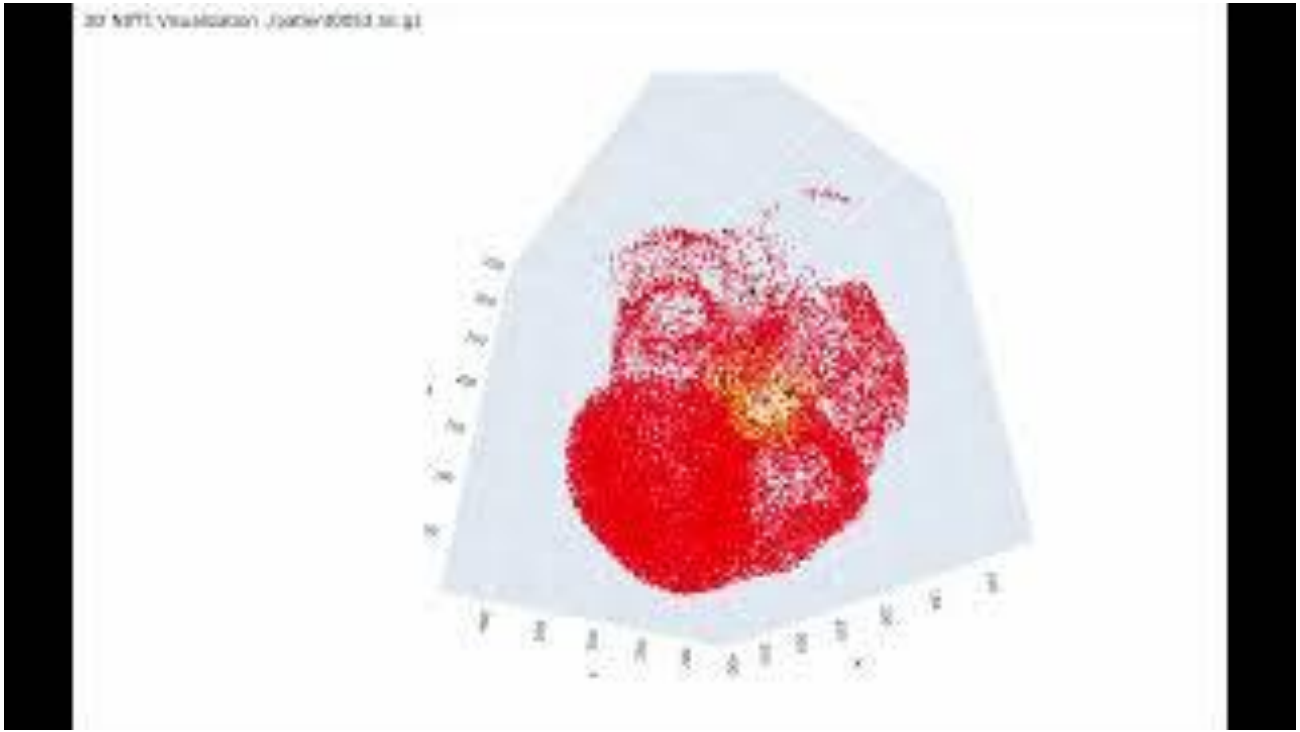


Demo Image – patient61





Demo Video



Future work

- **Impact of Pre-training Epochs:** Investigating the influence of the number of pre-training epochs on the final downstream performance.
- **Patch Size Analysis:** Evaluating the effect of different input patch sizes on this specific task's performance.
- **Data Normalization Comparison:** Comparing the results when using data processed with **Histogram Equalization** as input versus using the original, unnormalized data.

Thanks for listening